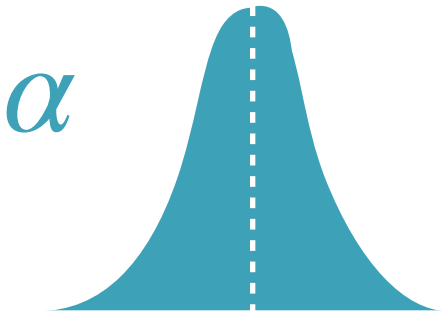
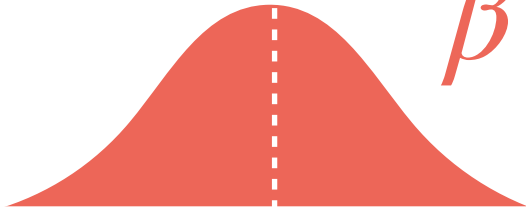


$$a \equiv \mathcal{N}(\mu_a, \sigma_a^2)$$

$$\beta = \mathcal{N}(\mu_{\beta}, \sigma_{\beta}^2)$$

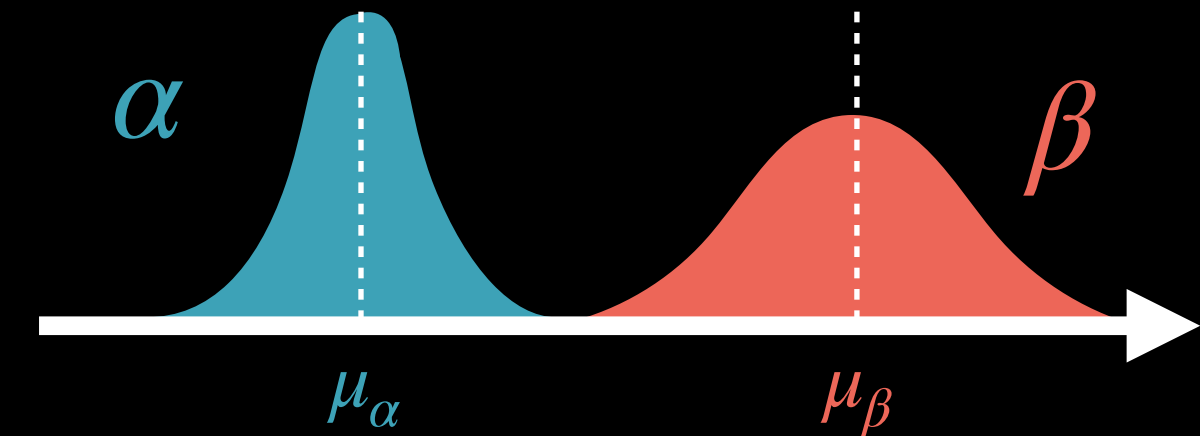
 α μ_{α}  β μ_{β}

• $(\mu_\alpha, \sigma_\alpha)$

• $(\mu_\beta, \sigma_\beta)$

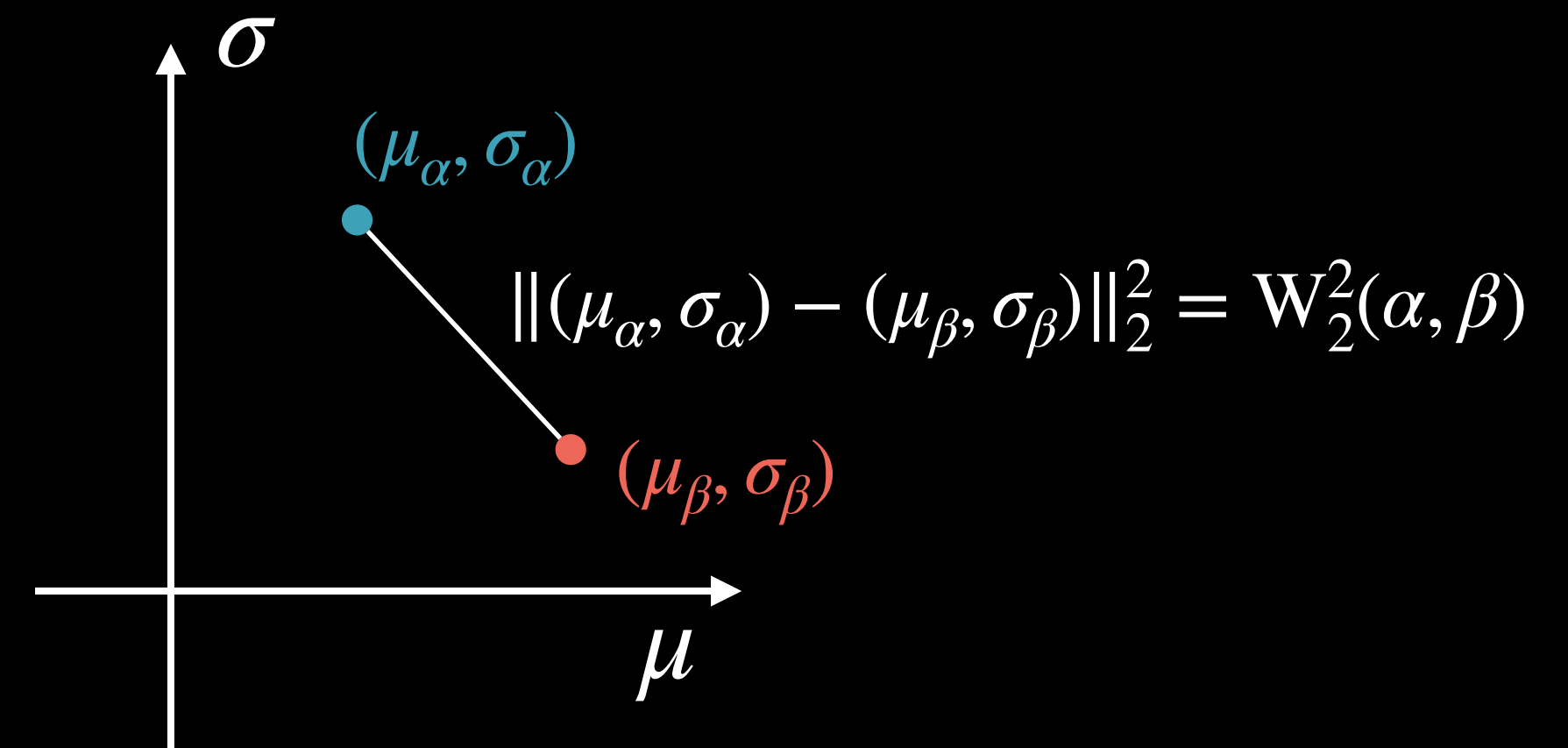
W_2 between 1D Gaussians

$$\alpha = \mathcal{N}(\mu_\alpha, \sigma_\alpha^2) \quad \beta = \mathcal{N}(\mu_\beta, \sigma_\beta^2)$$



Closed form solution:

$$W_2^2(\alpha, \beta) = (\mu_\alpha - \mu_\beta)^2 + (\sigma_\alpha - \sigma_\beta)^2$$



Wasserstein $\approx$ Euclidean distance in (mean, variance) space for Gaussians

W_2 between ND Gaussians

$$\alpha = \mathcal{N}(\mu_\alpha, \Sigma_\alpha)$$

$$\beta = \mathcal{N}(\mu_\beta, \Sigma_\beta)$$

